



Bertran Miquel Oliver

PhD Student in AI

- Barcelona, Spain
- bertranmiqueloliver@gmail.com

Links

- Personal webpage
- LinkedIn profile

Languages

- Catalan
- English
- French
- German
- Italian
- Spanish

Technical skills

- Python
- Git
- LaTeX
- R
- SQL
- C++
- C
- Mathematica

ML frameworks: PyTorch, PyTorch Geometric
HPC/tools: Linux, SLURM, WandB

Work experience

- Feb. 2024 - now **PhD Student**
BSC - Barcelona Supercomputing Center
- Sep. 2023 - Aug. 2025 **Part-time Lecturer**
UB - University of Barcelona
Teaching undergraduate Computer Science courses.
- June 2022 - June 2023 **Working student Data Analyst**
IFCO
Providing data analysis to worldwide customers.
Participating on a simulation project with Python.
- 2019 - 2021 **BI / Data roles**
Teladoc Health International; SDG
BI, databases, data modelling, Qlik and Tableau.

Academic formation

- April 2021 - August 2023 **MSc. Degree in Mathematics**
Minor in Data Science & Biomathematics
TUM - Technical University of Munich
Final degree thesis: *Mathematical Investigation into Genomic Order of Different Vertebrate Species* - Grade: 1.0
Relevant courses: *Introduction to Deep Learning, Machine Learning, Mathematical Models in Biology, Genetics*
- Sep. 2014 - June 2019 **BSc. Degree in Mathematics**
UB - University of Barcelona

Academic activities

- 2026 **LOGML Summer School**
Imperial College London
One-week summer school; awarded accommodation grant.

Publications

- 2026 **Effective Resistance Rewiring: A Simple Topological Correction for Over-Squashing**
Miquel-Oliver et al., PMLR, ICLR 2026 GRaM Workshop. arXiv:2603.11944.
- 2025 **Character-level Tokenizations as Powerful Inductive Biases for RNA Foundational Models**
Morales-Pastor et al., LMRL and AI4NA Workshops, ICLR 2025, Singapore. arXiv:2411.11808.
- 2025 **TopoBench: A Framework for Benchmarking Topological Deep Learning**
Telyatnikov et al., DMLR, ICLR Vienna. arXiv:2406.06642.